

Distilling Reliable PII Detection into a Deployable Redaction Model

Group: CS45-1

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01

Project Introduction

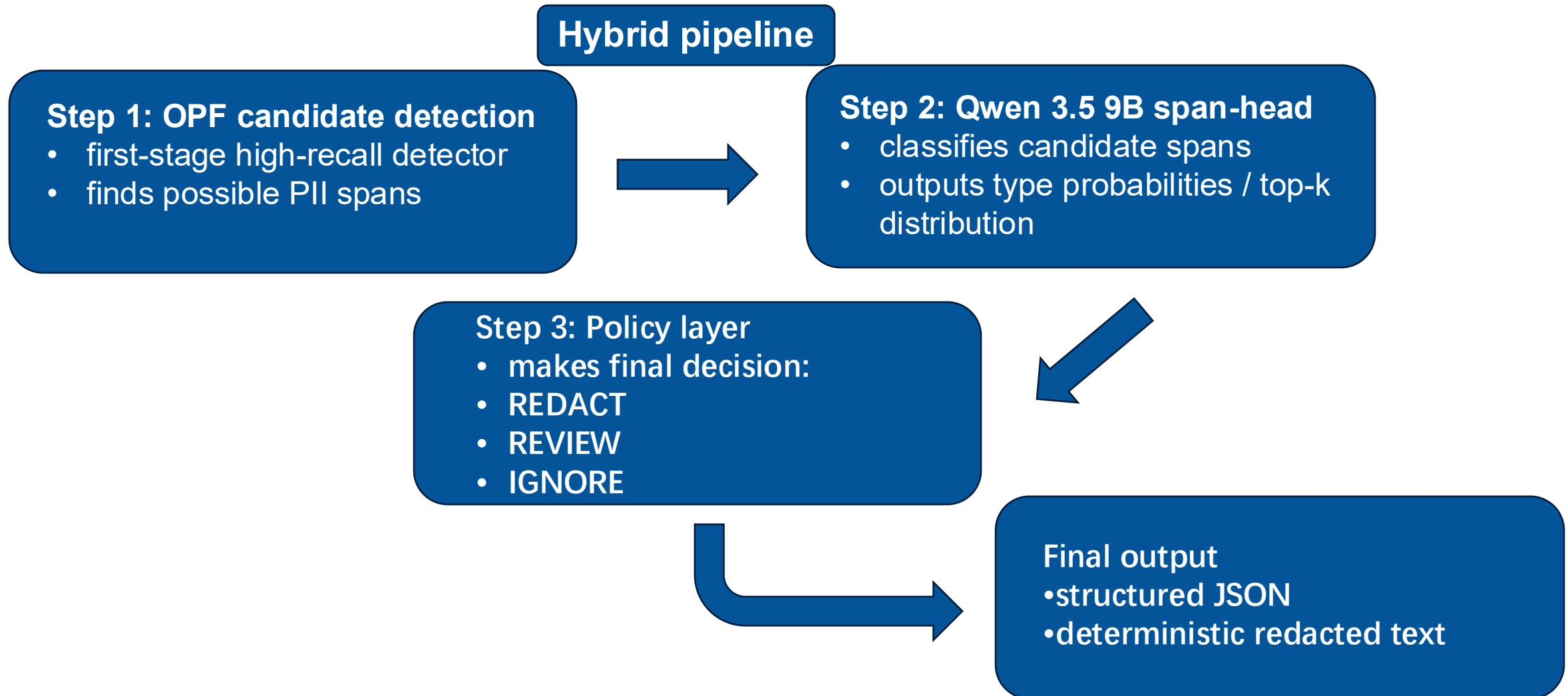
➤ Project Goal & Motivation

- Build a deployable Australian PII detection and redaction system
- Target sensitive entities such as:
 - PERSON
 - EMAIL
 - PHONE
 - TFN
 - Passport
 - Driver License

Our goal is to build a practical, structured, and privacy-safe system.



➤ Overall Implementation Idea



➤ Why Hybrid and What We Deliver

Why hybrid instead of one model

- **OPF**: fast scanning, high recall, stable boundaries
- **Qwen**: better semantic type judgement
- **Policy**: explainable and controllable decisions

Expected deliverables

- deployable service
- API + web demo
- support for text / image / PDF input
- stable output schema:
 - start / end
 - type
 - confidence
- three-way decision:
 - REDACT / REVIEW / IGNORE

02

Data Preparation

➤ Test Data for System Evaluation — Prepared samples used to verify the redaction pipeline

1

Coverage

TFN, Medicare, student ID, phone, email, address, name, DOB, bank details.

2

Sample Forms

Forms, emails, records, logs, payslip-style text, and narrative examples.

3

Hard Negatives

Invoice numbers, order IDs, reference numbers, and ordinary dates.

4

Evaluation Use

Test OPF detection, Qwen classification, and wrapper decisions.

Key message

Test data checks two things: whether the system finds real PII, and whether it avoids over-redacting ambiguous non-PII values.

OPF
detect spans

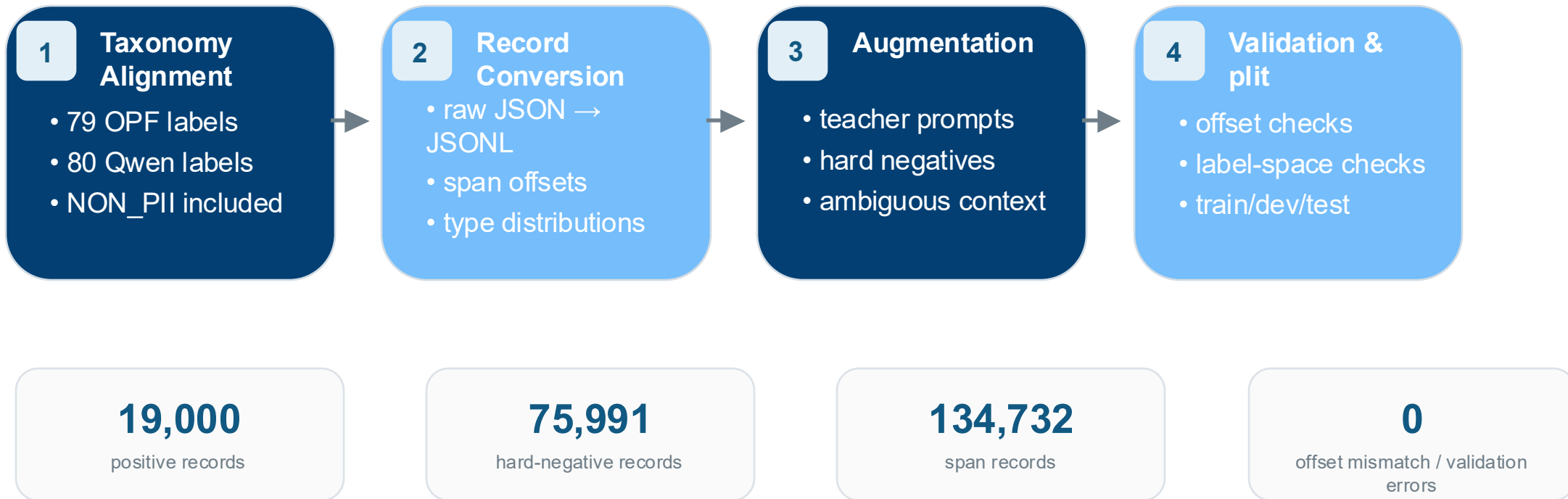


Qwen
classify type



Wrapper
decide action

➤ Data Preparation Logic - From raw AU PII records to test-ready span-distribution data



Key idea: test data must check both sensitivity detection and false-positive control.

➤ Test Data Record Format - Every sample is stored as machine-checkable text + span annotations

```
{
  "id": "sample_001",
  "text": "DOB: 04/05/1998 ...",
  "metadata": {
    "source_type": "form",
    "language": "en-AU"
  },
  "spans": [{
    "start": 5, "end": 15,
    "value": "04/05/1998",
    "top_type": "DATE_OF_BIRTH",
    "type_distribution": {
      "DATE_OF_BIRTH": 0.95,
      "NON_PII": 0.05
    }
  }]
}
```

Text + metadata

Context source such as form, email, record, log, payslip or narrative.

Span offsets

start / end / value are checked against the original text to avoid redaction boundary errors.

Probability distribution

Positive labels use smoothed distributions, while NON_PII is kept for ambiguous or negative cases.

Training weight

Teacher confidence becomes the training weight.

➤ Coverage Strategy: Realistic AU + University Context

Client-defined Taxonomy

- AU + University-specific PII
- 79 OPF labels
- 80 Qwen labels including NON_PII
- Focus: coverage, not redesigning labels

Scenario-based Samples

- Forms / emails / records
- Logs / chat text
- Payslip-style and narrative examples
- Text is the main input
- Image/PDF inputs are converted to text first

Hard Negatives

- Invoice / order / reference numbers
- Course codes / public URLs
- Placeholder emails / sandbox cards
- Ordinary dates / office phone lines
- Expected output: NON_PII or no span

Format logic: test real PII, ambiguous cases, and hard negatives.

➤ Quality Gates Before Evaluation — Prepared data must be valid before it is used by OPF or Qwen

1 Offset Alignment text[start:end] must equal the stored span value

2 Distribution Validity probabilities should sum to 1 and top_type must be argmax

3 Label-space Check no labels outside the 79/80-class taxonomy

4 Format Check rule candidates are stored for structured identifiers

5 Negative Control document-level hard negatives have empty spans

Evaluation-ready outputs

- OPF hard-label data: train/dev/test split
- Qwen span-head data: candidate span examples
- Wrapper smoke tests: real PII + hard negatives
- API/demo examples: structured JSON and redacted text

Validation result

0 offset mismatch / 0 validation errors in main stage-1 build

03

Model Selection and Training

➤ Selection Strategy & Architecture Construction

Core Strategy: Knowledge Distillation

“Large Teacher + Small
Student”
Soft Distillation framework

01. Teacher & Student Evaluation

Teacher: Qwen3.5-27B

Student: Qwen3.5-9B / 1.5B

Result: 1.5B achieved 174–188 ms response time

02. Native Multi-Modal Extension

Model: Qwen3.5-VL

Input: Text + Images + PDFs

Benefit: Replaces OCR-only workflow with layout-aware understanding

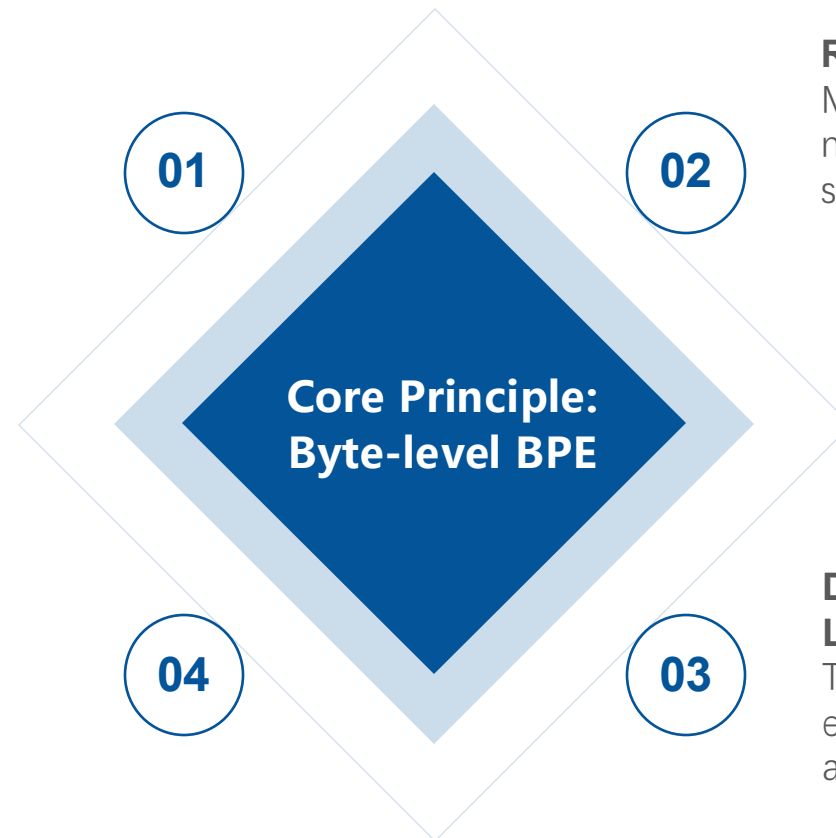
➤ Byte-level BPE Tokenizer

OOV Elimination & Format Compatibility

Natively compatible with mixed strings, plain digits, and complex AU identifiers (e.g., TFN, Medicare). Completely eliminates OOV (Out-Of-Vocabulary) errors and ensures encoding stability.

High Tolerance for OCR Noise

Remains reliable on garbled or noisy OCR outputs in multimodal pipelines.



Robustness in Blind Tests

Maintains stable encoding for noisy text and hard negative samples.

Distillation & Boundary Location

Teacher–Student alignment and enhances PII span boundary accuracy.

• OPF: Span Detection

Token-level sequence labelling > candidate PII spans > Qwen classification head

What OPF Does

Open-source token classifier
(OpenAI Privacy Filter)

Labels every token:
B-PERSON / I-PERSON / O / ...

Merges token labels into spans:
"John Smith" [11:21] → PERSON

Why Fine-Tune

Original OPF: generic PII only
(name, email, phone)

We need Australian types:
TFN Medicare Student ID
USI UAC ID WAM score
Bank account Passport

→ 79-class AU label space

Training Configuration

81,298 records
109,877 PII spans

3 epochs batch size 2
gradient accumulation 8
learning rate 1e-5

~2.3 hours on CUDA GPU
resulting checkpoint ~2.8 GB

• OPF Detection Results

Test set: 9,659 records (13,513 PII spans + 7,557 empty-span records)

0.979

Detection F1

0.984

Detection Recall

0.975

Detection Precision

Per-Class Span F1 (79 types)

47 classes F1 > 0.95

TFN 1.00 Medicare 1.00 Student ID 0.99
Passport 1.00 Mobile 1.00 DOB 0.96

7 classes F1 0.70 ~ 0.95

Email 0.90 Nationality 0.92 Payment Card 0.89

4 classes F1 < 0.70

Gender 0.36 Longitude 0.31 Latitude 0.52

Key Points

Format-stable AU identifiers:
near-perfect detection

Context-dependent types (Gender,
Lat/Long): weak no fixed pattern

Precision 0.975 on 7,557 empty-span
records low false-alarm rate

OPF is the high-recall first pass;
Qwen head refines the ambiguous cases

➤ Qwen Span-head Fine-tuning: Interface & Objective

This section starts after candidate spans are already available

Input to this stage

- Original text + surrounding context
- Candidate span offsets: start / end
- Span value to be typed
- Boundary detection is upstream, not here

Fine-tuning target

- 80-class probability distribution
- 79 PII labels + NON_PII
- Teacher-labelled soft targets
- Preserve ambiguity across similar labels

Output to wrapper policy

- Top-k type probabilities
- Calibrated confidence scores
- No final redaction decision yet
- Policy layer handles operational risk

Scope: soft-label training objective, calibration, and policy decision — not OPF detection or last_linear selection.

➤ Post-processing Policy: Probability → Action

The model predicts probabilities; the wrapper controls operational risk

Policy inputs

Top-1 probability
Top-k type distribution
NON_PII probability
Negative context signals
Uncertainty / risk score

1. Evidence gate

Reject weak PII evidence before risk scoring



2. Risk scoring

Use uncertainty and privacy risk to route cases



3. Decision

Produce REDACT / REVIEW / IGNORE

IGNORE

High NON_PII probability
or weak PII evidence

REVIEW

Potential PII, but
uncertainty remains high

REDACT

Strong PII evidence and
low uncertainty

Key message: probability is a model output; REDACT / REVIEW / IGNORE is a policy decision.

➤ Qwen Span Classification Head

What problem does it solve?

The Problem

OPF detects candidate spans but cannot reliably distinguish types.

"123456789" could be Student ID, TFN, Bank Account, or NON_PII depending on surrounding context.

The Solution

Freeze a Qwen3.5-9B backbone and train a tiny classification head on cached span embeddings.

No fine-tuning of the 9B model. Only the head is trained.

The Goal

80-class probability distribution per candidate span.

Output feeds into the policy layer for REDACT / REVIEW / IGNORE decisions.

➤ Qwen Span Classification Head

Why last_linear?

Head architecture comparison and selection rationale

Architecture	Dev NLL	Temperature	Parameters
mean_linear	0.124	1.165	~328K
first_linear	0.149	1.324	~328K
last_linear (selected)	0.085	1.005	~328K
concat_mlp	0.105	1.137	~5.4M

Why the last token?

Qwen is an autoregressive model: it reads text left to right. By the time it reaches the final token of a span like "John Smith", the hidden state at "Smith" already encodes information from "John" and the surrounding context.

The last token naturally aggregates evidence from the entire span.

Why a single linear layer?

concat_mlp has 16x more parameters but performs worse (dev NLL 0.105 vs 0.085).

For this task, representations from the frozen backbone are already linearly separable. Adding hidden layers introduces unnecessary overfitting.

Temperature 1.005 means raw logits require almost no calibration.

➤ Qwen Span Classification Head

Implementation & Results

From frozen backbone to deployed artefact

How it works

1. Freeze Qwen3.5-9B-Base (0 trainable params)
2. Cache span embeddings: mean, first, last
140,792 spans, 0 mapping failures
3. Train Linear(4096 to 80) on cached features
4. Apply Temperature Scaling ($T = 1.005$)
5. Deploy as single file (1.3 MB on disk)

Test Set Results

98.75%

Top-1 Accuracy

99.98%

Top-3 Accuracy

0.9755

Macro F1 (80 classes)

Impact

A 1.3 MB head replaces the need for a generative LLM at inference time: faster, cheaper, and more controllable than prompting a model for structured JSON. The lightweight head is version-controlled; only backbone checkpoints need GPU deployment.

➤ VLM for Multimodal Document Understanding

- Qwen3.5-9B-Base: process *image* + *text*
- Visual tokenization: visual tokens
- Grounded extraction: text fields with spatial regions, e.g. name, DOB, license number.
- Structured decoding: output format into JSON



OUTPUT

150075.4 ms

FILE DETAILS

File: driver_license.jpg · Text source: qwen_vl_ocr · Pages: 1 · Extracted text:
THE GOVERNMENT OF WESTERN AUSTRALIA DRIVER'S LICENCE WESTERN AUSTRALIA Department of Transport LICENCE
NUMBER 6437772 Given Name CITIZEN, Family Name JOHN PETER Middle Name 20 BROWN STREET EAST PERTH WA
6004 EXPIRY DATE 17 Jan 2019 CLASS C SAMPLE DATE OF BIRTH 18 Dec 1965 18121965

ENTITIES

THE GOVERNMENT OF WESTERN AUSTRALIA DRIVER'S LICENCE WESTERN AUSTRALIA Department of Transport
LICENCE NUMBER 6437772 Given Name CITIZEN, Family Name **JOHN PETER** PERSON Middle Name 20 BROWN
STREET EAST PERTH WA 6004 EXPIRY DATE 17 Jan 2019 CLASS C SAMPLE DATE OF BIRTH **18 Dec 1965**
DATE_OF_BIRTH 18121965

AUTO-REDACTED

THE GOVERNMENT OF WESTERN AUSTRALIA DRIVER'S LICENCE WESTERN AUSTRALIA Department of Transport LICENCE
NUMBER 6437772 Given Name CITIZEN, Family Name JOHN PETER Middle Name 20 BROWN STREET EAST PERTH WA
6004 EXPIRY DATE 17 Jan 2019 CLASS C SAMPLE DATE OF BIRTH [DATE_OF_BIRTH] 18121965

04

Demo Website

➤ Demo Website

- API layer connects frontend input with backend PII detection.

Input: text

*Input:
PDF/image*

REDACT PII detection & redaction opf-hard-79-qwen9b-hn-spanhead-v1 Ready

INPUT

TEXT 354 / 3000

Matter reference DXPH-3405. This memo concerns O. Wang and is circulated on a need-to-know basis. Born 1969-12-10; the file further records that gender recorded as Non-binary. Residing at Unit 35, 109 Pacific Cres, Tennant Creek NT 0810; supplementary information indicates mobile on 0464180938. Records additionally show contactable at oliver@proton.me.

FILE

Choose a PDF, image, or text file

Run → Clear

OUTPUT 76759.2 ms

ENTITIES

Matter reference DXPH-3405. This memo concerns **O. Wang** PERSON and is circulated on a need-to-know basis. Born **1969-12-10** DATE_OF_BIRTH; the file further records that gender recorded as **Non-binary** GENDER. Residing at **Unit 35, 109 Pacific Cres, Tennant Creek NT 0810** ADDRESS; supplementary information indicates mobile on **0464180938** PHONE. Records additionally show contactable at **oliver@proton.me** EMAIL.

AUTO-REDACTED

Matter reference DXPH-3405. This memo concerns [PERSON] and is circulated on a need-to-know basis. Born [DATE_OF_BIRTH]; the file further records that gender recorded as Non-binary. Residing at [ADDRESS]; supplementary information indicates mobile on [PHONE]. Records additionally show contactable at [EMAIL].

5 redact · 1 review

VALUE	TYPE	CONFIDENTIALITY	PROB	RISK	DECISION	TOP-K DISTRIBUTION
O. Wang	PERSON	Protected	95.6%	LOW	REDACT	PERSON 95.6% SOCIAL_MEDIA_ACCOUNT 0.0% SANCTIONS 0.0% EMAIL_ADDRESS 0.0% PASSPORT_START_DATE 0.0%
1969-12-10	DATE_OF_BIRTH	Protected	97.1%	LOW	REDACT	DATE_OF_BIRTH 97.1% PASSPORT_START_DATE 0.0% CREDIT_CARD_EXPIRY 0.0% PASSPORT_EXPIRY 0.0% MOBILE 0.0%
Non-binary	GENDER	Protected	100%	LOW	REVIEW	GENDER 100.0%
Unit 35, 109 Pacific Cres, Tennan...	ADDRESS	Protected	97.1%	LOW	REDACT	ADDRESS 97.1% MOBILE 0.0% UAC_ID 0.0% DATE_OF_BIRTH 0.0% AU_TFN 0.0%
0464180938	PHONE	Protected	95.6%	LOW	REDACT	MOBILE 95.6% ADDRESS 0.1% EMAIL_ADDRESS 0.0% WORK_PHONE 0.0% DATE_OF_BIRTH 0.0%
oliver@proton.me	EMAIL	Protected	96.2%	LOW	REDACT	EMAIL_ADDRESS 96.2% WORK_EMAIL 0.0% GEOLOCATION_INFORMATION 0.0% MOBILE 0.0% PERSON 0.0%

*Output:
Detected entities +
redacted text*

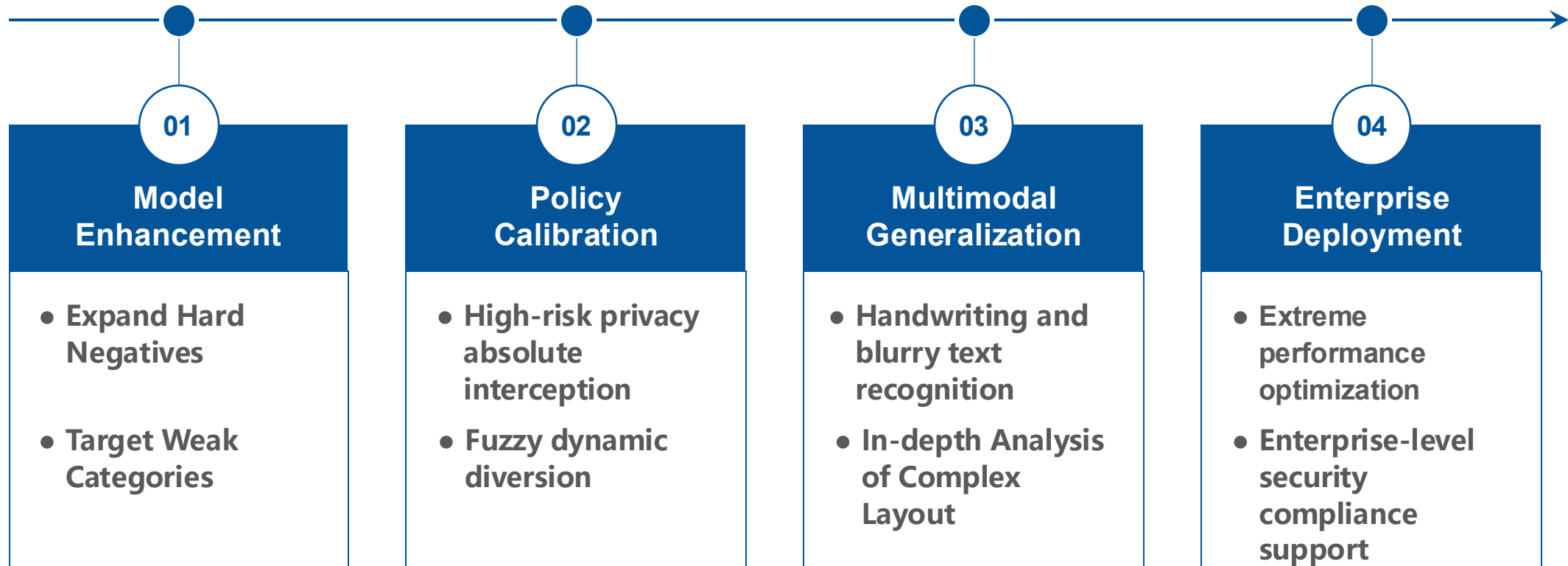
*Risk: combined
probability & privacy
level*

*Top-K Distribution:
top candidate labels
and their probabilities*

05

Reflection and Future Outlook

➤ Future Outlook





THANKS

Applicant: CS45-1 GROUP